Dental occlusion, labiodental articulation, and sound change: Evidence from a biomechanically diverse human speech dataset

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Introduction

- Hockett (1985) proposed that labiodental fricatives developed relatively recently in human history due to changes in how humans' upper and lower teeth come together.
- Blasi et al. (2019) used biomechanical modeling to argue that labiodental fricatives [f v] require less effort in agricultural societies than in hunter-gatherer societies.
- We examine these claims using pre-operative speech recordings of jaw surgery patients with a wide range of occlusal relationships.

Figure 1: Blasi et al. (2019). Left: Skull with edge-to-edge bite; Middle: simulation of labiodental with overjet and overbite; Right: simulation of labiodental with edge-to-edge bite.

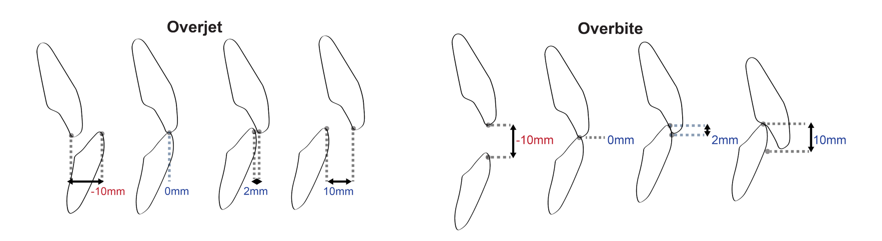

Figure 2: Overjet (OJ) and overbite (OB). 0mm: edge-to-edge occlusion, common in hunter-gatherer societies. 2mm: normal occlusion in present-day humans. -10 and 10mm: significant OJ and OB malocclusion.

Variation in overjet and overbite

- Overjet is the *horizontal* distance between the upper and lower incisors.
- Overbite is the amount of *vertical* overlap between incisors.
- In agricultural societies, +2mm of overbite (OB) and overjet (OJ) became typical (Proffit et al. 2018) (Figure 2).
- Blasi et al. (2019)'s biomechanical modeling compared with +2mm of OB and OJ with an edge-to-edge bite (0mm of OB and OJ) and found that [f v] require less effort and are more likely to occur by accident at +2mm OJ and OB than at 0mm OB and OJ.
- People with significantly abnormal OB and OJ ranges receive surgical correction and often endorse speech distortions as a reason for surgery (Silva et al. 2026, Bode et al. 2023), but /f v/ are not sounds typically reported as challenging for surgical patients.
- Excessive negative OJ has been associated with higher distortion of other consonants /s ʃ t tʃ/ and surgical correction to normal OJ values trends with improved speech clarity (Tran et al. 2024, Silva et al. 2026, Figures 3).
- 2mm is tiny compared to the range of OB and OJ values in the pre-surgical population.

Previous studies

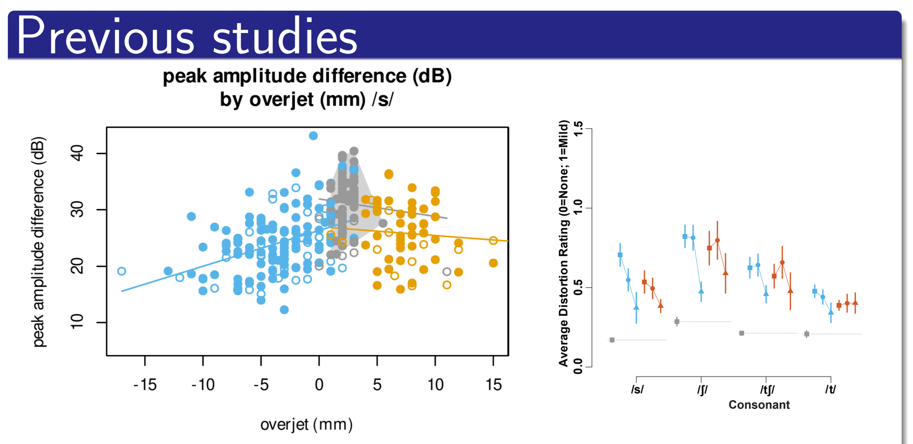

Figure 3: Left: /s/ noise amplitude by degree of overjet (Tran et al. 2024); Right: change in distortion ratings over 1 year for selected consonants from a Class III surgical population (Silva et al. 2026)

Speakers and Target Words

- 243 American-English speaking surgical patients and 51 Class I controls were included (Table 1).
- OJ values ranged from -17 to 18mm and OB values ranged from -9 to 12mm.
- Tokens of the words "**f**our", "**f**i**v**e", "se**v**en", "**p**urr", and "co**p**" were selected. 3024 tokens were measured in total.

Patient group	OJ (mm)	OB (mm)	Count
Control	∼ 2	∼ 2	51
Class I AOB	∼ 2	< 0	5
Class II	> 4	∼ 2	50
Class II AOB	> 4	< 0	19
Class III	< 0	∼ 2	124
Class III AOB	< 0	< 0	45

Table 1: Speaker counts by and typical overjet and overbite by group.

Data preparation methods

- Recordings were segmented at the phone level using the Montreal Forced Aligner (McAuliffe et al. 2017) with the english_us_arpa models and hand-corrected in Praat (Boersma and Weenink 2007).

Distortion rating methods

- A mix of 12 lay and expert listeners auditorily rated each token as normal (0), mildly distorted (1), moderately distorted (2), or severely distorted (3). Listeners were blinded to speaker and tokens were randomized.
- 3 raters with phonetic training transcribed each token in IPA.
- Speakers were classified as 'distorted' if their average distortion value fell outside of the 95th percentile of the controls (0.75).

Acoustic analysis methods

- Multitaper spectra were created from 20ms windows with preemphasis ($\alpha = 0.99$) using the `spectRum` package (Reidy 2013) in R (R Core Team 2015).
- Spectra were selected from noise peak for /f v/ and release for /p/.
- /f v/ are characterized by the main spectral peak amplitude relative to low-frequency trough and /p/ is characterized by release amplitude relative to closure.

Distortion rating results

- Distortion is rare across *all* patient groups (Figure 4): no robust difference between Controls and edge-to-edge bite (Class III mild) or more extreme underbite.

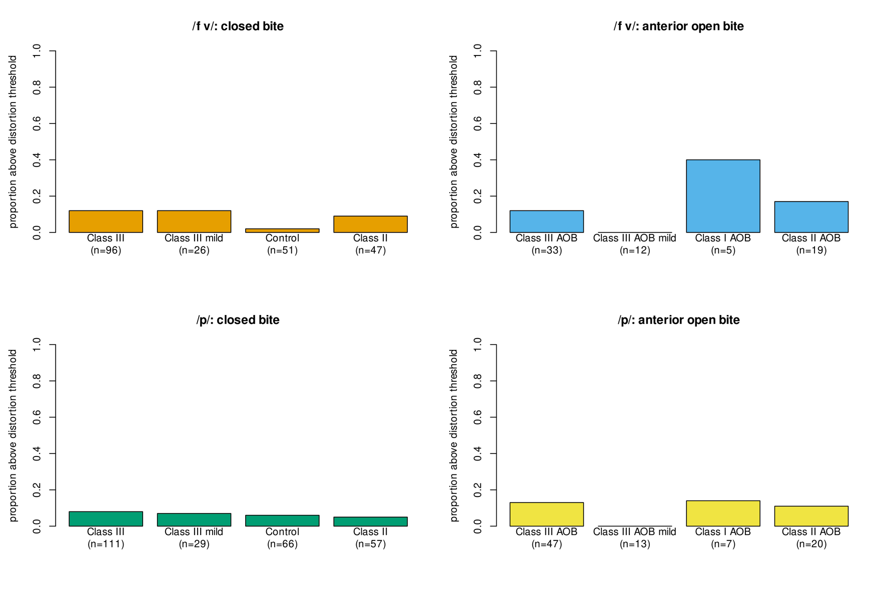

Figure 4: Proportion of patients over distortion threshold.

Acoustic results

- Distorted /f v/ have reduced spectral peak amplitude and slightly reduced peak frequency (Figure 5 left).
- Distorted /f v/ are not abundant in edge-to-edge or more extreme Class III regions, but mostly absent from positive overbite (Class II) region (Figure 5 right).
- /p/ distortion is less closely related to spectral shape, no evidence of accidental [f] being more frequent in positive overbite (Class II) region (Figure 5 bottom).
- No significant relationship found between overjet and distortion or acoustic measures of /f v/ 6.
- Extreme positive or negative overbite is associated with distortion in /f v p/ and reduced peak frequency in /f/.
- /p/ release amplitude is the only measure correlated with overjet.

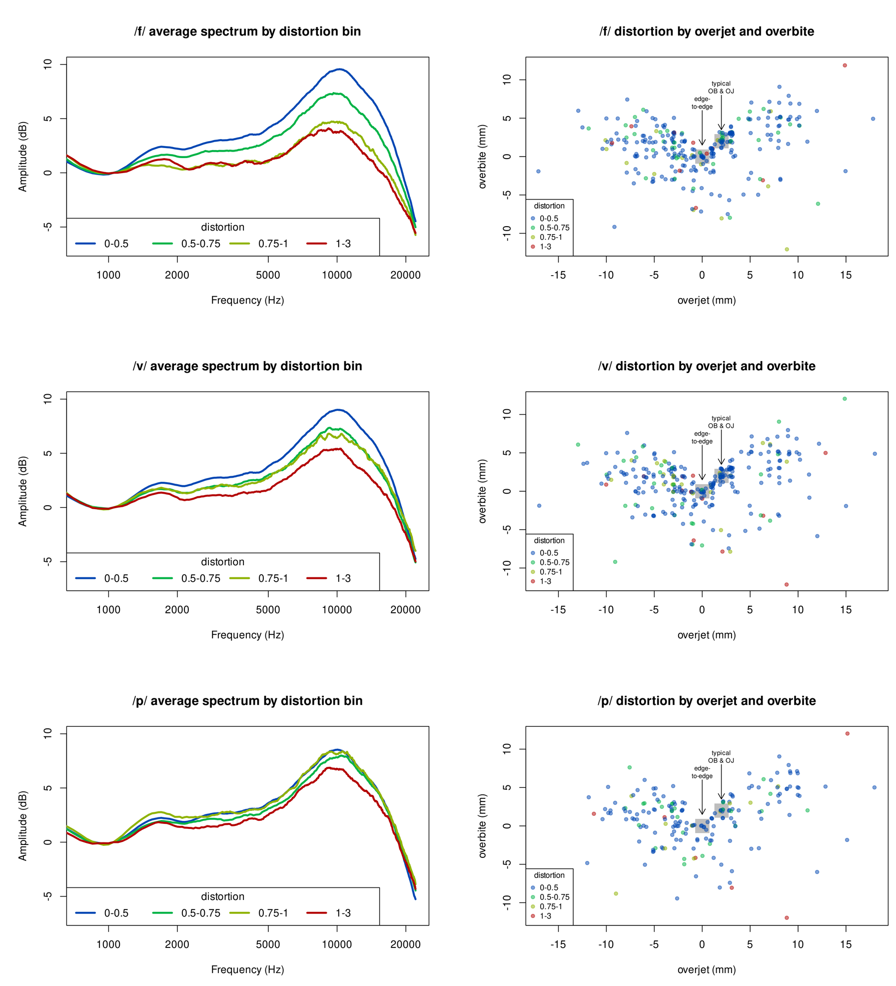

Figure 5: Left: average spectra by distortion bin (fricatives rated as distorted are less noisy); Right: distortion bin by OB and OJ.

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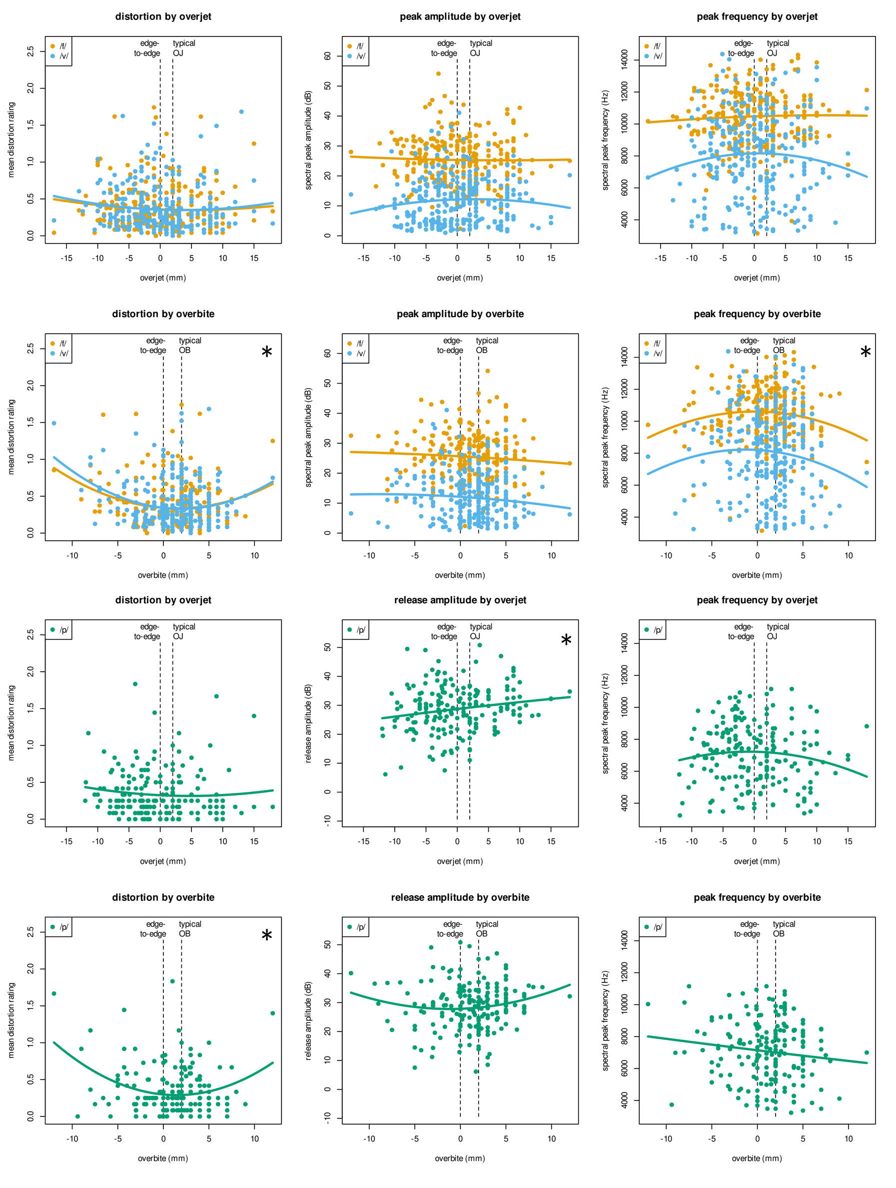

Figure 6: Distortion and acoustic measures related to OJ and OB with 2nd degree polynomial fits. No significant effect of OJ on labiodentals.

Discussion and Conclusions

- Biomechanical modeling and human word-reading performance yielded different estimates of the role of occlusion in labiodental articulation.
- We did not detect large differences in distortion or acoustic measures of /f v/ between humans with typical bites, edge-to-edge bites, or more extreme underbites.
- Most speakers with extreme OB and OJ values can produce labiodentals that are acoustically similar to those of controls.
- We also don't see /p/ pronounced as [f] in patients with positive overjet, so we did not find support for the idea that accidental labiodental articulation is more likely at more positive overjet values.

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